

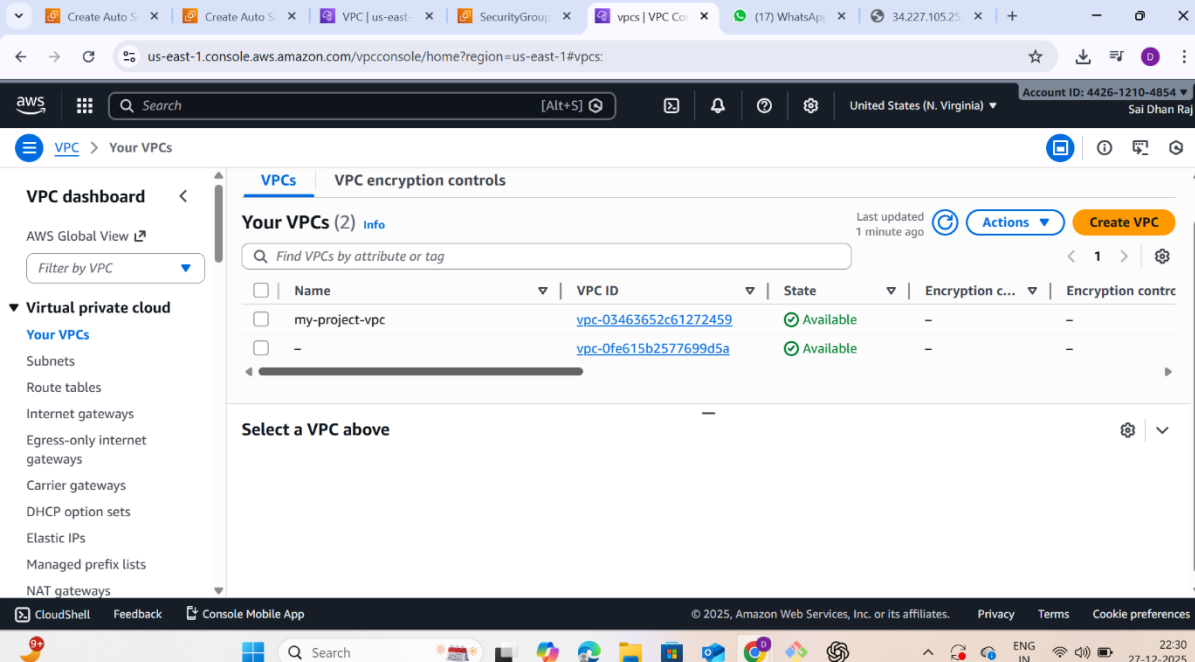


3-Tier Architecture

For Web Application In AWS

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Batch No :145

creating vpc

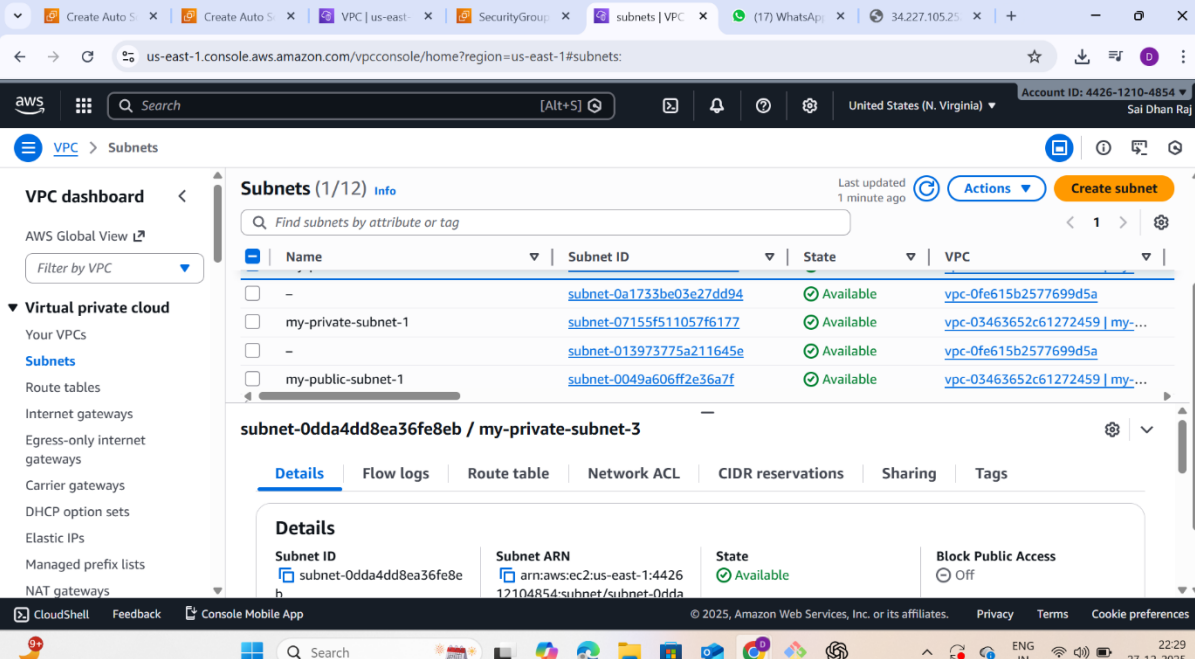


The screenshot shows the AWS Management Console for the region us-east-1. The left sidebar displays the 'VPC dashboard' with a list of 'Your VPCs'. The main content area shows 'Your VPCs (2)' with a table listing two VPCs:

Name	VPC ID	State	Encryption c...	Encryption contr...
my-project-vpc	vpc-03463652c61272459	Available	-	-
-	vpc-0fe615b2577699d5a	Available	-	-

Below the table, there is a section titled 'Select a VPC above' with a dropdown menu.

Creating subnets: 2 public ,2 private, 2 RDS Subnets



The screenshot shows the AWS Management Console for the region us-east-1, specifically the 'Subnets' page. The left sidebar displays the 'VPC dashboard' with a list of 'Your VPCs'. The main content area shows 'Subnets (1/12)' with a table listing four subnets:

Name	Subnet ID	State	VPC
-	subnet-0a1733be03e27dd94	Available	vpc-0fe615b2577699d5a
my-private-subnet-1	subnet-07155f511057f6177	Available	vpc-03463652c61272459 my-...
-	subnet-013973775a211645e	Available	vpc-0fe615b2577699d5a
my-public-subnet-1	subnet-0049a606ff2e36a7f	Available	vpc-03463652c61272459 my-...

Below the table, there is a section titled 'subnet-0dda4dd8ea36fe8eb / my-private-subnet-3' with tabs for 'Details', 'Flow logs', 'Route table', 'Network ACL', 'CIDR reservations', 'Sharing', and 'Tags'. The 'Details' tab is selected, showing the following information:

Subnet ID	Subnet ARN	State	Block Public Access
subnet-0dda4dd8ea36fe8eb	arn:aws:ec2:us-east-1:442612104854:subnet/subnet-0dda4dd8ea36fe8eb	Available	Off

create internet gateway

The screenshot shows the AWS Management Console for the 'us-east-1' region. The main content area is titled 'Internet gateways (1)'. It features a search bar and a table with the following data:

Name	Internet gateway ID	State	VPC ID
my-project-internetgateway	igw-019c03834df5adf0a	Attached	vpc-03463652c61272459 my-

The left sidebar shows the 'VPC dashboard' with a search bar and a list of resources: Your VPCs, Subnets, Route tables, Internet gateways (selected), Egress-only internet gateways, Carrier gateways, DHCP option sets, Elastic IPs, Managed prefix lists, and NAT gateways. The top navigation bar shows the user is in the 'us-east-1' region and has an account ID of 4426-1210-4854.

create nat gateway

The screenshot shows the AWS Management Console for the 'us-east-1' region. The main content area is titled 'NAT gateways (1)'. It features a search bar and a table with the following data:

Name	NAT gateway ID	Connectivity...	State	State message	Availabi
my-public-natgateway	nat-026513b9c077c18fe	Public	Available	-	Zonal

The left sidebar shows the 'VPC dashboard' with a search bar and a list of resources: Route tables, Internet gateways, Egress-only internet gateways, Carrier gateways, DHCP option sets, Elastic IPs, Managed prefix lists, NAT gateways (selected), Peering connections, Route servers, Security (Network ACLs, Security groups), and PrivateLink and Lattice (Getting started). The top navigation bar shows the user is in the 'us-east-1' region and has an account ID of 4426-1210-4854.

create routetables

Route tables (1/3)

Name	Route table ID	Explicit subnet associations	Edge associations	Main
my-routetable-project	rtb-04b30a4ef4c08a51c	-	-	No
-	rtb-03bfe19541bd76571	-	-	Yes
-	rtb-0262f332678a7abf4	-	-	Yes

rtb-04b30a4ef4c08a51c / my-routetable-project

Details

Route table ID	Main	Explicit subnet associations	Edge associations
rtb-04b30a4ef4c08a51c	No	-	-

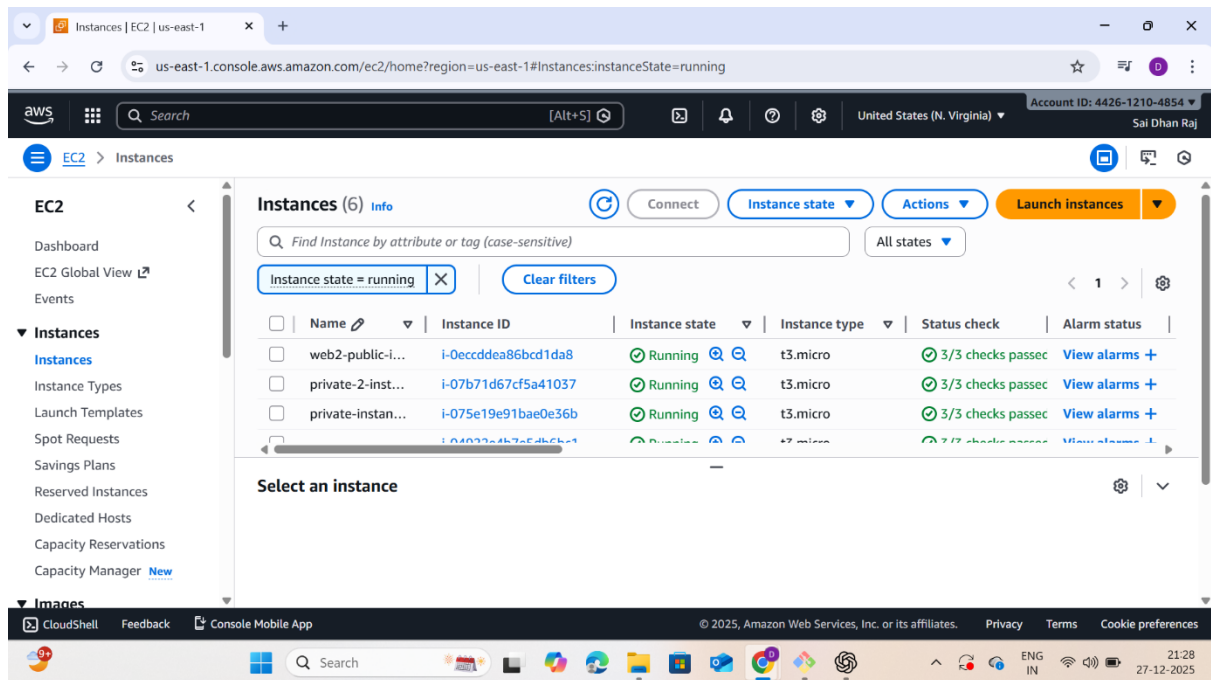
Create security groups

Security Groups (6)

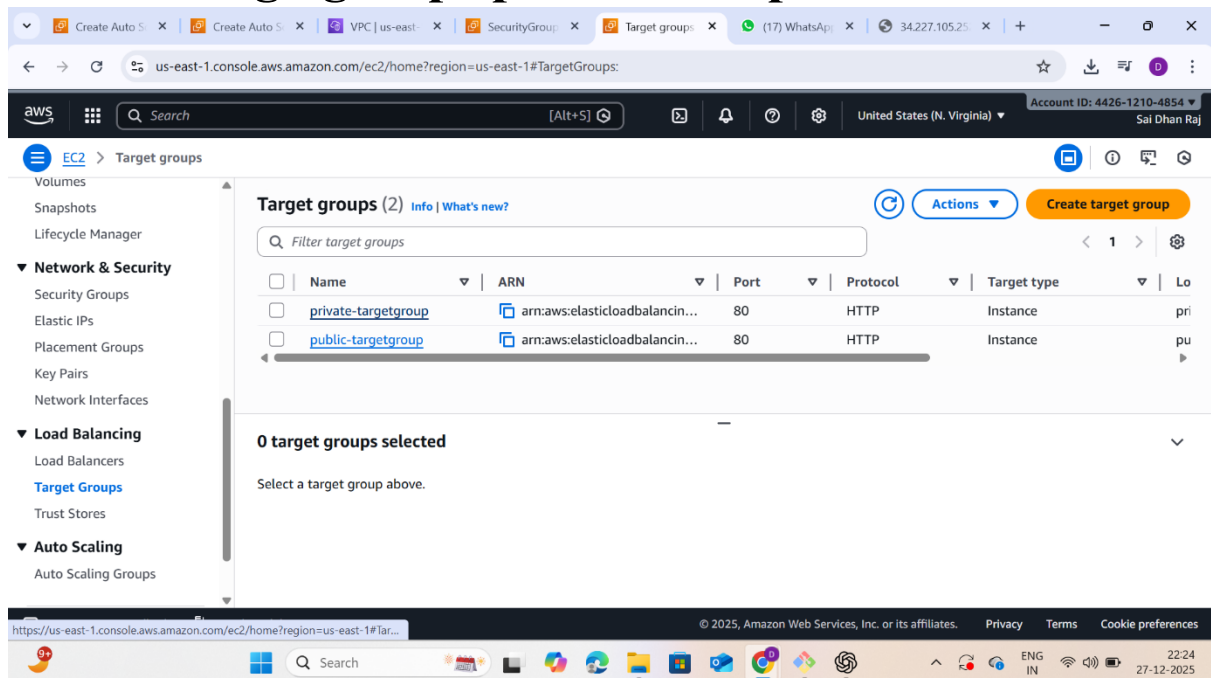
Name	Security group ID	Security group name	VPC ID
-	sg-07cc9b49867ff726b	my-autoscaling	vpc-0fe615b2577699d5
-	sg-091a97b5bd5b0cfc0	launch-wizard-1	vpc-0fe615b2577699d5
-	sg-094479446a28ad5cc	launch-wizard-2	vpc-0fe615b2577699d5
-	sg-04d0c8dc39ef545e1	my-project-sg	vpc-03463652c6127245

Select a security group

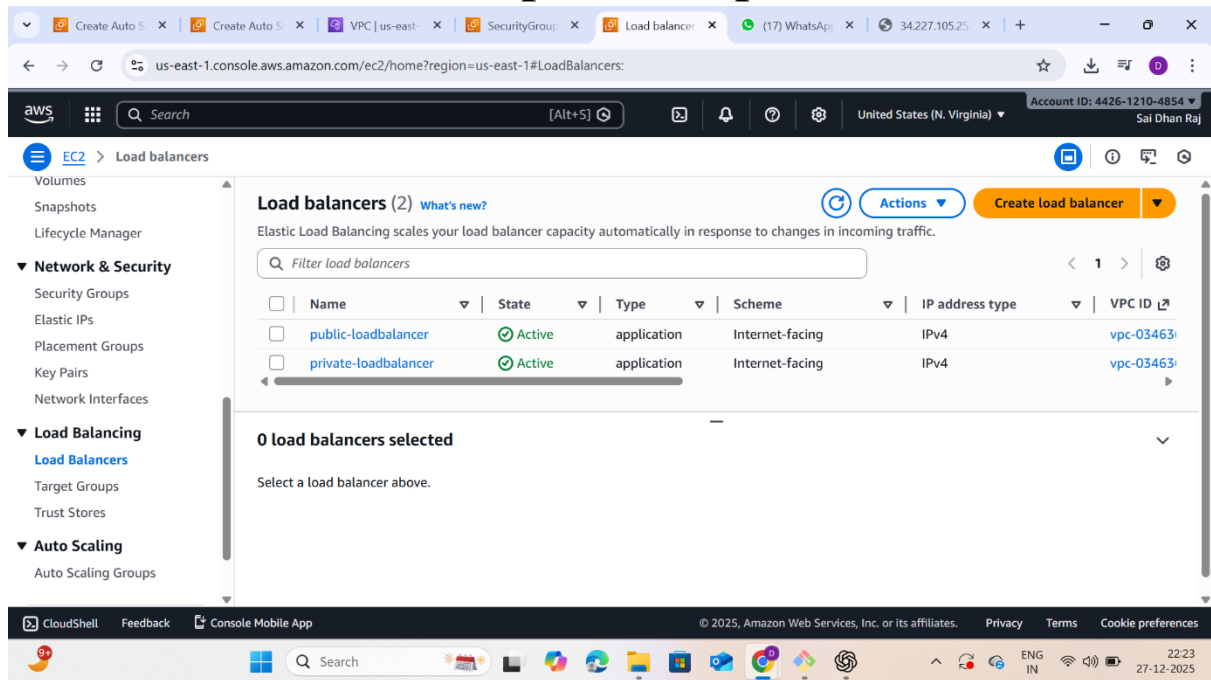
Create ec2 instance: 2 public and 2 private instance



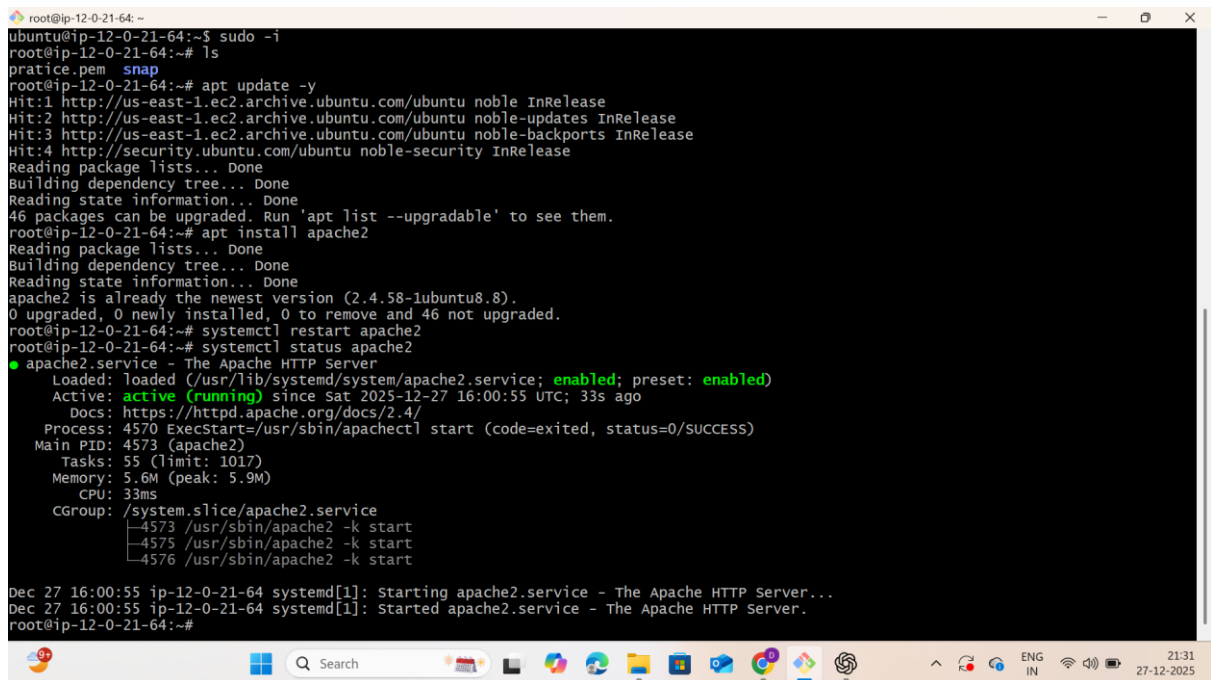
Create Targetgroups private and public



create load balancers: public & private

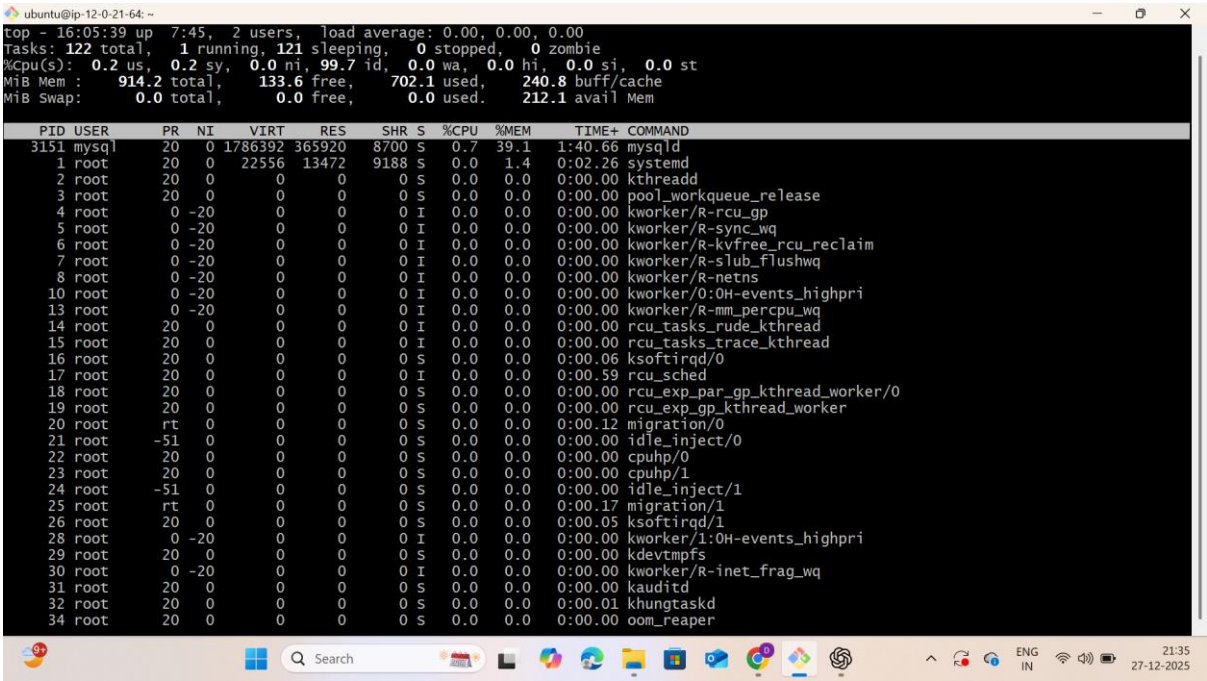


connect in servers

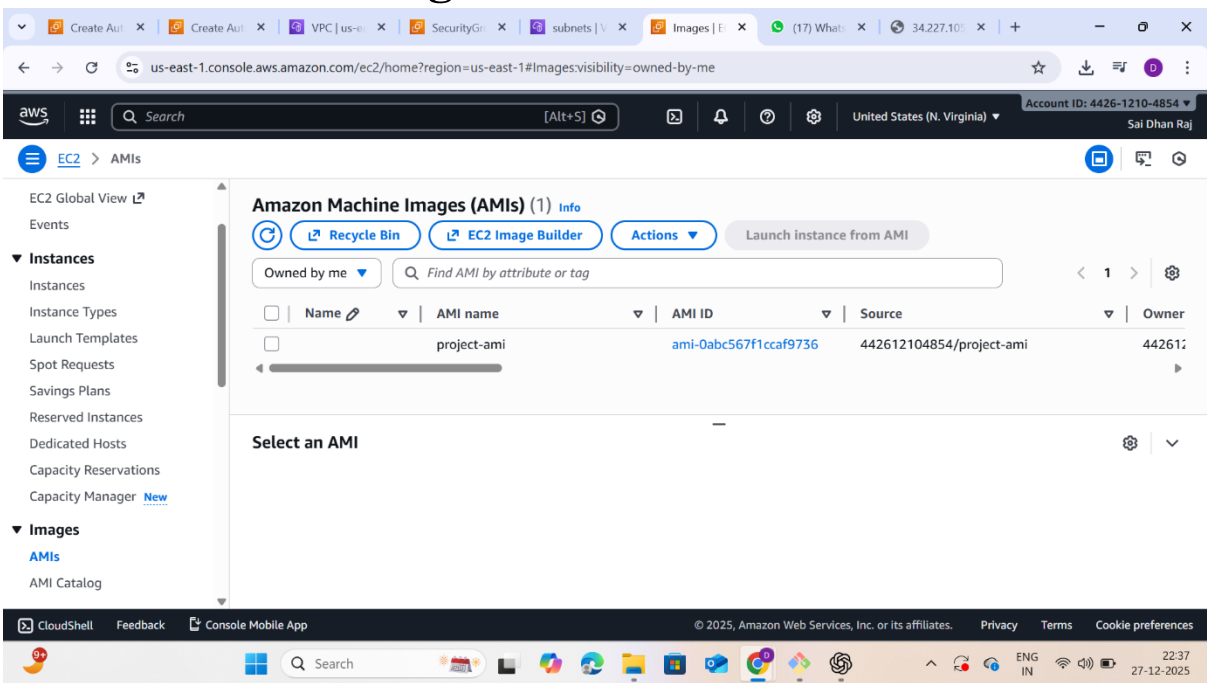


inst

server runing state



create AMI through ec2instance



create autoscaling

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Auto Scaling groups' page is active, displaying a list of existing groups. The 'private-autoscaling' group is using the 'AUTOSCALING-TEMP' launch template and has 2 instances. The 'public-autoscaling' group is using the 'my-autoscaling-template' launch template and has 0 instances. Both groups are in a 'Deleting' state. The page includes a search bar, a 'Create Auto Scaling group' button, and a sidebar with navigation options like Network & Security, Load Balancing, and Auto Scaling.

Name	Launch template/configuration	Version	Instances	Status	Desired Capacity
private-autoscaling	AUTOSCALING-TEMP	Version Default	2	Deleting	0
public-autoscaling	my-autoscaling-template	Version Default	0	Deleting	0

create aurora & RDS

The screenshot shows the AWS Management Console for the 'us-east-1' region. The 'Subnet groups' page is active, displaying a list of existing groups. The 'project-subnet-group' is using the 'allow' VPC and has a status of 'Complete'. The page includes a search bar, a 'Create DB subnet group' button, and a sidebar with navigation options like Aurora and RDS.

Name	Description	Status	VPC
project-subnet-group	allow	Complete	vpc-03463652c61272459

create tables in mysql

```
root@ip-12-0-21-64: ~
-> AC
mysql> clear
mysql> use devops;
Database changed
mysql> CREATE TABLE users (
->   id INT AUTO_INCREMENT PRIMARY KEY,
->   name VARCHAR(50),
->   email VARCHAR(100),
->   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.02 sec)

mysql> show tables;
+-----+
| Tables_in_devops |
+-----+
| users            |
+-----+
1 row in set (0.01 sec)

mysql> INSERT INTO users (name, email)
-> VALUES
-> ('Dhanraj', 'dhanraj@gmail.com'),
-> ('Admin', 'admin@gmail.com');
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> select * from users;
+----+-----+-----+-----+
| id | name  | email          | created_at |
+----+-----+-----+-----+
| 1  | Dhanraj | dhanraj@gmail.com | 2025-12-27 13:01:34 |
| 2  | Admin  | admin@gmail.com  | 2025-12-27 13:01:34 |
+----+-----+-----+-----+
2 rows in set (0.01 sec)

mysql> Read from remote host ec2-35-175-116-75.compute-1.amazonaws.com: Connection reset by peer
```

```
root@ip-12-0-21-64: ~
Server version: 8.0.43 Source distribution
copyright (c) 2000, 2025, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or
its affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current
input statement.

mysql> show databases;
+-----+
| Database |
+-----+
| devops   |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
5 rows in set (0.00 sec)

mysql> create databases aws;
ERROR 1064 (42000): You have an error in your SQL syntax; check
the manual that corresponds to your MySQL server version for
the right syntax to use near 'databases aws' at line 1
mysql> CREATE DATABASE aws;
ERROR 1290 (HY000): The MySQL server is running with the --read-
only option so it cannot execute this statement
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mysql>
```